24.

① $$\lim_{x\to+\infty}\frac{\boxed{\int_0^x e^{t^2}dt}}{e^{x^2}}$$ 求导 ∞ ∞

$F(x)=\int_0^x f(t)\,dt$

$F'(x)=f(x)$

$$=\lim_{x\to+\infty}\frac{e^{x^2}}{2xe^{x^2}}$$

$$=\lim_{x\to+\infty}\frac{1}{2x}=0$$

② $\infty - \infty$

$$\lim_{x\to+\infty}\sqrt{4x^2+7x}-2x$$

$$=\lim_{x\to+\infty}\frac{(\sqrt{4x^2+7x}-2x)(\sqrt{4x^2+7x}+2x)}{\sqrt{4x^2+7x}+2x}$$

$$=\lim_{x\to+\infty}\frac{(4x^2+7x)-(2x)^2}{\sqrt{4x^2+7x}+2x}$$

$$=\lim_{x\to+\infty}\frac{7x}{\underbrace{\sqrt{4x^2+7x}}_{一次}+\underbrace{2x}_{一次}}$$

一次 多项式 多项式

$$= \lim_{x \to +\infty} \frac{7}{\sqrt{4} + 2} = \frac{7}{4}$$

$$= \lim_{x \to +\infty} \frac{7}{\frac{\sqrt{4x^2+7x}}{x} + 2}$$

$$= \lim_{x \to +\infty} \frac{7}{\sqrt{4 + 7 \cdot \frac{1}{x}} + 2}$$

$$\frac{1}{x} \to 0$$

$$= \frac{7}{\sqrt{4} + 2} = \frac{7}{4}.$$

$$\lim_{x \to 0} \frac{\frac{d}{dx} \frac{\sin x}{x}}{x} \to \text{求导}$$

$$\frac{d}{dx}\left(\frac{\sin t}{x}\right) = ?$$

$$= \left(\frac{\sin x}{x}\right)' = \frac{x\cos x - \sin x}{x^2}$$

$$\lim_{x\to 0} \frac{\dfrac{x\cos x - \sin x}{x^2}}{x}$$

$$= \lim_{x\to 0} \frac{x\cos x - \sin x}{x^3}$$

$$= \lim_{x\to 0} \frac{\cancel{\cos x} - x\sin x - \cancel{\cos x}}{3x^2}$$

$$= -\frac{1}{3} \lim_{x\to 0} \frac{x\sin x}{x^2}$$

$$= -\frac{1}{3}$$

25.

① $y = x^{\sin x}$

① $\ln y = \sin x \ln x$

$$\frac{dy}{y} = \left(\cos x \ln x + \frac{\sin x}{x}\right) dx$$

$$\Rightarrow \frac{dy}{dx} = x^{\sin x} \cdot \left(\cos x \ln x + \frac{\sin x}{x}\right).$$

② $y = e^{\sin x \ln x}$

$$y' = e^{\sin x \ln x} \cdot (\sin x \ln x)'$$

$$= e^{\sin x \ln x} \cdot \left(\cos x \ln x + \frac{\sin x}{x}\right)$$

④ $f(x) = x^2 \sin x$

$$f^{(9)}(0)$$

$$f^{(9)}(x) = \sum_{k=0}^{9} C_9^k \cdot (x^2)^{(k)} \cdot (\sin x)^{(9-k)}$$

$$= C_9^0 (x^2)^{(0)} \cdot (\sin x)^{(9)} + C_9^1 (x^2)^{(1)} \cdot (\sin x)^{(8)} + C_9^2 (x^2)^{(2)} \cdot (\sin x)^{(7)}$$

$$= 1 \cdot x^2 \cdot \cos x + 9 \cdot (2x) \cdot (\sin x) + 36 \cdot 2 \cdot (-\cos x)$$

$$f^{(9)}(0) = -72$$

$$\frac{f^{(9)}(0)}{9!} = x^9 \text{的系数}$$

$$f(x) = x^2 \cdot \left(x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \frac{1}{7!}x^7 + \cdots\right)$$

$$= x^3 - \frac{1}{3!}x^5 + \frac{1}{5!}x^7 - \frac{1}{7!}x^9 + \cdots$$

$$\frac{f^{(9)}(0)}{9!} = -\frac{1}{7!}$$

$$\Rightarrow f^{(9)}(0) = -\frac{9!}{7!}$$

$$= -(9 \times 8) = -72$$

26. $\int \arccos(2x)\, dx$

$2x$

$t = 2x$

$= \frac{1}{2}\int \arccos t \, dt$

$= \frac{1}{2}\left(t\arccos t - \int t \, d(\arccos t)\right)$

$= \frac{1}{2}\left(t\arccos t - \int t\cdot\left(-\frac{1}{\sqrt{1-t^2}}\right) dt\right.$

$= \frac{1}{2}\left(t\arccos t + \int \frac{t}{\sqrt{1-t^2}}\, dt\right.$

$u = t^2 = 4x^2$

$= \frac{1}{2} t\arccos t + \frac{1}{2}\int \frac{\frac{1}{2}\, du}{\sqrt{1-u}}$

$= \frac{1}{2} t\arccos t + \frac{1}{4}\int \frac{1}{\sqrt{1-u}}\, du$

$\left((1-u)^{\frac{1}{2}}\right)' = \frac{1}{2}\cdot(1-u)^{-\frac{1}{2}}\cdot(-1)$

$\Rightarrow (1-u)^{-\frac{1}{2}} = \left(-2(1-u)^{\frac{1}{2}}\right)'$

$= \frac{1}{2} t\arccos t + \frac{1}{4}\left(-2(1-u)^{\frac{1}{2}}\right) + C$

$$= \frac{1}{2} t \arccos t - \frac{1}{2}(1-u)^{\frac{1}{2}} + C$$

$$= x \arccos 2x - \frac{1}{2}(1-4x^2)^{\frac{1}{2}} + C$$

$$\int \arccos(2x)\, dx$$

$2x = \cos t$

$2dx = -\sin t \, dt$

$t = \arccos 2x$

$$= \int t \left(-\frac{1}{2}\sin t\right) dt$$

$$= -\frac{1}{2}\int t \sin t \, dt$$

$$= -\frac{1}{2}\int t \, d(-\cos t)$$

$$= -\frac{1}{2}\left(t \cdot (-\cos t) - \int (-\cos t)\, dt\right)$$

$$= \frac{1}{2} t\cos t - \frac{1}{2}\int \cos t \, dt$$

$= \frac{1}{2} t\cos t - \frac{1}{2}\sin t + C$

$= \frac{1}{2}\arccos 2x \cdot 2x - \boxed{\frac{1}{2}\sin(\arccos(2x))}$

$+ C$

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② $\int_0^5 |x^2 - 5x + 4| \, dx$

or $x > 4$ $x^2 - 5x + 4 > 0$
$x < 1$

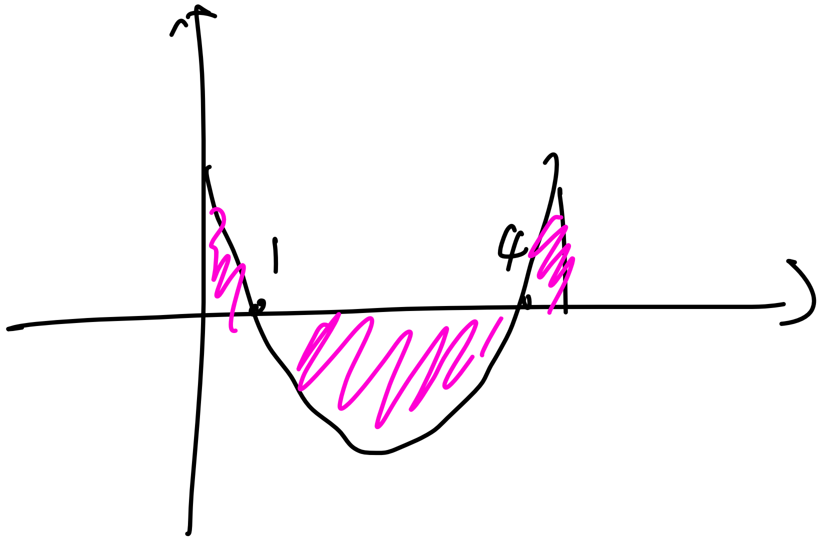
$1 < x < 4$ $x^2 - 5x + 4 < 0$

$=$

$= \int_0^1 (x^2 - 5x + 4) \, dx$

$+ \int_1^4 -(x^2 - 5x + 4) \, dx$

$+ \int_4^5 (x^2 - 5x + 4) \, dx$



③ $$\int_0^1 \frac{dx}{\sqrt{x}+\sqrt[3]{x}}$$

$x = t^6$ $\qquad \sqrt{x} = t^3$

$dx = 6t^5\,dt$ $\qquad \sqrt[3]{x} = t^2$

$$\int_0^1 \frac{6t^5\,dt}{t^2+t^3} = \int_0^1 \frac{6t^3}{1+t}\,dt$$